

① First Class:

-Def. A polytope, P , is the convex hull of a set of vertices in $\mathbb{R}^N$. i.e. $v_1, \dots, v_k \in \mathbb{R}^N \Rightarrow \left\{ \sum_{i=1}^k \lambda_i v_i \mid 0 \leq \lambda_i \leq 1 \right\}$

-Exs:



"Simplicial": all proper facets are simplexes Δ , $\triangle$, etc.

Face vector: $f = (f_0, f_1, \dots, f_{d-1})$, $f_i = \#$ of i -dimensional faces

Q: Given a vector f , when is there an associated polytope?

[Simpler: can we answer for what f there is an associated simplicial polytope?]

McMullen: Better to associate an h -vector to P .

→ Inductive definition (induction on faces of P)

→ For simplicial P , $h(P) = (h_0, \dots, h_d)$ are coeffs. of

$$h(P, t) = (t-1)^d + f_0(t-1)^{d-1} + \dots + f_{d-1}$$

e.g. $P = \Delta_n$ a n -diml simplex

$$\Rightarrow h(\Delta_n, t) = t^d + t^{d-1} + \dots + 1$$

$P = \text{cube}$

$$\Rightarrow h(P, t) = t^3 + 5t^2 + 5t + 1$$

Conjecture (McMullen) / Thm (Stanley): $h \in \mathbb{Z}_{\geq 0}^{d+1}$ is an h -vector for a d -diml

simplicial polytope P iff

- ① $h_i = h_{d-i} \quad \forall i$
- ② $\exists$ graded $\mathbb{Q}$ -alg. R with $R_0 = \mathbb{Q}$, R generated in deg 1,
 $\dim_{\mathbb{Q}} R_i = h_i - h_{i-1} \quad \forall 1 \leq i \leq \lfloor d/2 \rfloor$

Cor. P : simplicial $\Rightarrow h(P)$ is unimodal

Stanley's idea: $P \longleftrightarrow$ "fans" $\longleftrightarrow$ toric variety $X_P / \mathbb{C}$

Thm: P : simplicial $\Rightarrow h(P, t) = \sum_{k=0}^d \dim H^{2k}(X_P, \mathbb{Q}) \cdot t^k$
 $\& H^{\text{odd}}(X_P, \mathbb{Q}) = 0$

When P is not simplicial, X_P is not " $\mathbb{Q}$ -smooth" & Stanley's theorem fails. But there is a fix!

② Thm (Stanley): For general rational $\mathbb{P}$,

$$h(\mathbb{P}, t) = \sum_{k=0}^d \underbrace{\dim \mathrm{IH}^{2k}(X_{\mathbb{P}}, \mathbb{Q})}_{\text{"Intersection homology"}} \cdot t^k$$

~ We will return later in the course to prove this. ~

What is "Intersection (co)homology"?

A: provides a good (co)homology theory for singular varieties
 "good": we can do intersection theory with it.

Intersection Theory for smooth varieties:

X : smooth, oriented n -manifold, $H_n(X; \mathbb{Z}) \cong \mathbb{Z} \ni [X]$ fundamental class

Thm (Poincaré Duality). $\cap [X]: H^i(X, \mathbb{Z}) \rightarrow H_{n-i}(X, \mathbb{Z})$

So with rational coeffs:

$$\begin{array}{ccc} H^i(X, \mathbb{Q}) & \xrightarrow{\sim} & H_{n-i}(X, \mathbb{Q}) \\ \parallel \cong & \text{universal coeffs. thm} & \parallel \cong \\ H_i(X, \mathbb{Q})^* & & H^{n-i}(X, \mathbb{Q})^* \end{array}$$

Cor: $h_i = \dim_{\mathbb{Q}} H_i(X, \mathbb{Q}) = \dim_{\mathbb{Q}} H^{n-i}(X, \mathbb{Q})$ is a symmetric/palindromic sequence.

Better: can specify the isomorphism $H^{n-i}(X, \mathbb{Q}) \cong H^{n+i}(X, \mathbb{Q})$

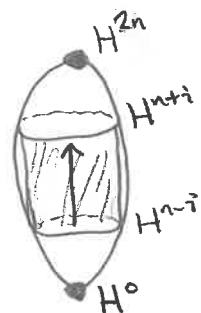
Assume $X \subseteq \mathbb{P}_{\mathbb{C}}^N$ is a complex, projective variety.

Choose a generic hyperplane $H \subset \mathbb{P}^N$

$$\leadsto [X \cap H] \in H_{2n-2}(X, \mathbb{Z}) \xLeftrightarrow{\text{P.D.}} [H] \in H^2(X, \mathbb{Z})$$

Thm (Hard Lefschetz). $H^{n-i}(X, \mathbb{Q}) \xrightarrow[\sim]{-v[H]^i} H^{n+i}(X, \mathbb{Q})$ is isom.

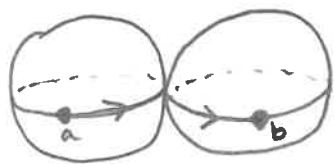
↳ Rmk on how "hard" Hard Lefschetz is: hard enough Lefschetz didn't prove it in his 1924 paper; Hodge did in 1941!



Cor: h_i are also unimodal.

③ This fails for singular spaces

Ex: $X = \mathbb{P}^1 \cup \mathbb{P}^1 \subseteq \mathbb{P}^2$



By Mayer-Vietoris:

$$H_i(X, \mathbb{Q}) = \begin{cases} \mathbb{Q} & i=0 \\ 0 & i=1 \\ \mathbb{Q} \oplus \mathbb{Q} & i=2 \end{cases} \quad \text{no Poincaré duality}$$

$H^i(X, \mathbb{Q})^*$

What is going on? a & b want to be distinct 0-cycles, but they are connected by a 1-cocycle through the singularity.

IH will exclude this cocycle:

$$IH^i(X, \mathbb{Q})^* \cong IH_i(X, \mathbb{Q}) = \begin{cases} \mathbb{Q} \oplus \mathbb{Q} & i=0 \\ 0 & i=1 \\ \mathbb{Q} \oplus \mathbb{Q} & i=2 \end{cases}$$

But what if I only care about cohomology? IH has a sheafy analogue called perverse sheaves.

cohomology sheaf of a proper map $\in \left\{ \begin{array}{l} \text{perverse} \\ \text{sheaves} \end{array} \right\}$

The category of perverse sheaves has atoms called

"Intersection Cohomology" (IC) sheaves.

Thm [Decomposition Thm]. $f: X \rightarrow B$ a proper map between smooth varieties, then $Rf_* \mathbb{Q}_X$ is a semisimple perverse sheaf, i.e. it is a direct sum of IC sheaves.

$\Rightarrow$ If you know the supports of summands, you can go from general fibers to smooth fibers!

Underlies Springer theory, proof of Fundamental Lemma, Kazhdan-Lusztig theory, proof of P=W conjecture, ... many many more!

(3.5)

Recap of some algebraic topology:Chains: $\Delta^n = n$ -simplex $C_n(X)$ = free abelian group generated by maps $\Delta^n \rightarrow X$

$$\partial_n(\sigma) = \sum_{i=0}^n (-1)^i \cdot (\sigma \circ d_i) \quad \text{for } d_i: \Delta^{n-1} \hookrightarrow \Delta^n \text{ the } i\text{th face.}$$

$$H_i(X) = H_i(\dots \rightarrow C_n(X) \rightarrow C_{n-1}(X) \rightarrow \dots)$$

Rmk: Can replace $C_n(X)$ with set of (possibly infinite) sums $\sum c_i \sigma_i$ s.t. every cpt $K \subset X$ intersects only finitely many σ_i . In this case, the complex computes Borel-Moore homology, a sort of "homology w/cpt supports".

e.g. $\sigma_i: \Delta^1 \rightarrow \mathbb{R}$ by the interval $[i, i+1]$

$$\sum_{i \in \mathbb{Z}} \sigma_i \in C_1^{\text{BM}}(\mathbb{R})$$

Exercise: $\sum_{i \in \mathbb{Z}} \sigma_i$ generates $H_1^{\text{BM}}(\mathbb{R}) \cong \mathbb{Z}$.

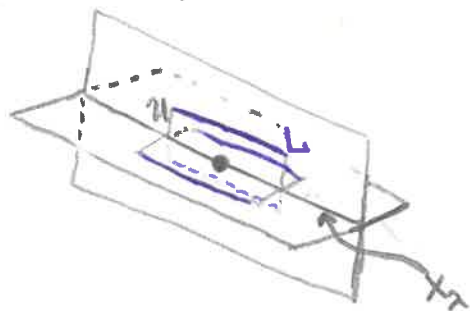
Whitney Stratifications (CORRECTED): $n = \dim_{\mathbb{R}}(X)$

$X = \bigsqcup_{\lambda} X_{\lambda}$ is a stratification if $\overline{X_{\lambda}}$ is a union of X_{μ} for some set of μ .

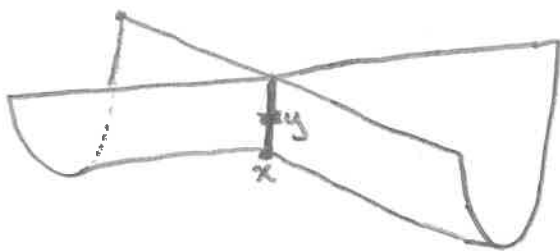
This stratification is a Whitney stratification if

$\forall \lambda$, there is a stratified space $L = \bigsqcup_{\mu} L_{\mu}$ of $\dim(L) = n - \dim(X_{\lambda}) - 1$ s.t. $\forall x \in X_{\lambda}$, $\exists$ neighborhood $\mathcal{U} \ni x$ s.t. $\phi: \mathcal{U} \xrightarrow{\text{homeo.}} \mathbb{R}^{\dim(X_{\lambda})} \times \text{cone}(L)$, where
 $\text{cone}(L) = L \times [0, 1] / L \times \{0\}$

$\& \phi^{-1}(L_{\mu}) = X_{\mu} \cap \mathcal{U} \& \phi|_{\mathcal{U} \cap X_{\lambda}}: \mathcal{U} \cap X_{\lambda} \xrightarrow{\sim} \mathbb{R}^{\dim(X_{\lambda})} \times \{0\}$



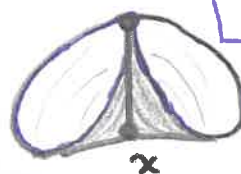
Ex: Whitney Umbrella: $\{x^2 = zy^2\} \subseteq \mathbb{C}^3$



At y:

$y \in \mathcal{U}: \mathcal{U} \cap X_{\lambda} \approx \mathbb{R} \times \text{cone}(L)$

At x:



$L = S^1 \vee S^1 = (S^1 \vee S^1 \setminus \{pt\}) \sqcup \{pt\}$
 (for real picture)

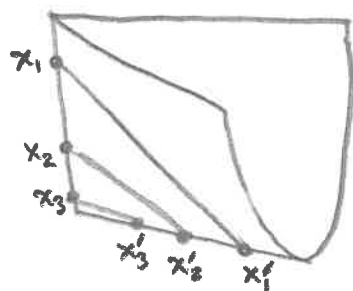
These are equivalent to the Whitney conditions:

(a) If $x_i \rightarrow x$ for $x_i \in X_2$ & $x \in X_M$, then

$$T_x X_M \subseteq \lim_{i \rightarrow \infty} T_{x_i} X_2 \text{ if this limit exists}$$

(b) If $x_i \in X_2$ & $x'_i \in X_M$ both go to $x \in X_2$, then $\lim_{i \rightarrow \infty} [\text{Secant line } (x_i, x'_i)] \subseteq \lim_{i \rightarrow \infty} T_{x_i} X_2$ if limits exist.

Ex: for Whitney umbrella, can take



} limit of secant lines is horizontal, but $\lim_{i \rightarrow \infty} T_{x_i} X_2$ is vertical line only

So we need extra strata.

(Complex) Link of Whitney Umbrellas:

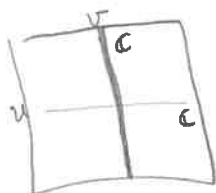
$$\pi: \mathbb{C}^2 \rightarrow X$$

$$(u,v) \mapsto (uv, v, u^2)$$

compatible w/ contraction

$$\lambda \cdot (u,v) = (\lambda u, \lambda^2 v) \quad \& \quad \lambda \cdot (x,y,z)$$

$$= (\lambda^3 x, \lambda^2 y, \lambda^2 z)$$



glues $\pm u$ coords.

$$\text{Link at } (0,0) \text{ is } X \setminus \{0\} / G_m \cong \mathbb{C}^2 \setminus \{0\} / \left(\begin{array}{l} (u,0) = (-u,0) \forall u \\ \lambda(u,v) = (u,v) \forall \lambda \neq 0, v \end{array} \right)$$

$$\cong S^3 / \left(\begin{array}{l} (u,0) = (-u,0) \\ \text{on an unknotted} \\ S^1 \subset S^3 \end{array} \right)$$

Links of Isolated Singularities:

Thm (Milnor fibers). $f: \mathbb{C}^{n+1} \rightarrow \mathbb{C}$ s.t. $f(0) = 0$, f : holomorphic

Let $B_\varepsilon = \{z \in \mathbb{C}^{n+1} \mid \|z\| \leq \varepsilon\}$ be a small ball @ $0 \in \mathbb{C}^{n+1}$

Then $f^{-1}(0) \cap B_\varepsilon \underset{\text{homeo.}}{\approx} \text{cone}(L)$ for $L \underset{\mu}{\approx} \bigvee S^n$.
(homotopic)

$$\text{for } \mu = \dim_{\mathbb{C}} \frac{\mathbb{C}[x_0, \dots, x_{n+1}]}{(\frac{\partial f}{\partial x_0}, \dots, \frac{\partial f}{\partial x_{n+1}})}$$

Ex: $f: \mathbb{C}^2 \rightarrow \mathbb{C}$ has $\mu = \dim_{\mathbb{C}} \frac{\mathbb{C}[x,y]}{(2y, -3x^2)} = \dim_{\mathbb{C}} S_{\text{ann}\{1,x\}} = 2$
 $(x,y) \mapsto y^2 - x^3$

$$L \approx S^1 \vee S^1$$

Complex Link of $X = \{y^2 = x^3\} \subseteq \mathbb{C}^2$:

$$\pi: \mathbb{C} \rightarrow X$$

$$z \mapsto (z^2, z^3)$$

Write $z = re^{i\theta}$ & fix r .

$\Rightarrow (r^2 e^{2i\theta}, r^3 e^{3i\theta})$ traverses 1 circle 3 x's & the other twice.

$\Rightarrow$ torus knot in S^3 .

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LECTURE 2:

Intersection Homology: X : singular $\mathbb{C}$ -variety.

O: Sings. are

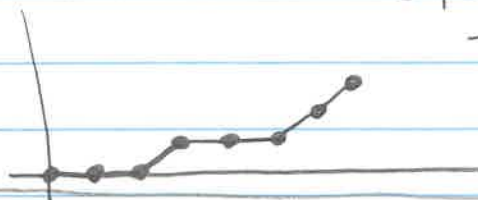
interesting

e.g. $x^n = y^m$

I: Whitney Stratifications

 $X = \coprod_{\lambda} X_{\lambda}$ stratification (ie. X_{λ} conn'd, locally clsd, smooth)
 $\overline{X_{\lambda}} = \bigcup_{\mu} X_{\mu}$ We ask X_{λ} is "equisingular", i.e. $\forall x \in X_{\lambda}, \exists$ ngbd $U \ni x$
s.t. $(X_{\lambda} \cap U) \times V \cong U$ (id on $X_{\lambda} \cap U$) ("local normal triviality")Ex: Whitney umbrella $\{x^2 = zy^2\} \subseteq \mathbb{C}^3$. $(u,v) \mapsto (uv, v, u^2)$ $X_{\text{reg}} \cup X_{\text{sing}}$ not good enough! $X_{\text{reg}} \cup (X_{\text{sing}} \setminus \{0\}) \cup \{0\}$ $y^2 + x^3 - t^2 x^2 = 0$: Whitney Condition: $X_{\mu} \subset \overline{X_{\lambda}} \Rightarrow \forall a_i \in X_{\lambda}, b_i \in X_{\mu}$ converging to $c \in X_{\mu} \Rightarrow \lim_{i \rightarrow \infty} (\text{secants } a_i - b_i) \in \lim_{i \rightarrow \infty} T_{c_i} X$ Intersection Homology: to recover P.D. / intersection theory
need to restrict what cycles we deal with.Def: A perversity is a fn $\overline{p}: [0, \dim X] \cap \mathbb{Z} \rightarrow \mathbb{N}$
s.t. $p_{\alpha} := \overline{p}(\alpha)$ satisfies $\begin{cases} p_0 = p_1 = p_2 = 0 \\ p_{\alpha} \leq p_{\alpha+1} \leq p_{\alpha+1} + 1, \alpha \geq 2 \end{cases}$

real dimension



slopes 0 or 1 everywhere.

e.g. upper middle perversity: $(0, 0, 0, 1, 1, 2, 2, \dots, \frac{n}{2}-1)$ lower middle: $(0, 0, 0, 0, 1, 1, 2, 2, \dots, \frac{n}{2}-1, \frac{n}{2}-1)$ $X = X_n \supset X_{n-1} \supset \dots \supset X_1 \supset X_0 \supset X_{-1} = \emptyset$ ($\dim X_i \leq i$)An i -chain $\xi \in C_i(X)$ is $\overline{p}$ -allowed if $\dim(\xi \cap X_{n-\alpha}) \leq i - \alpha + p_{\alpha}$ Note: $\xi \not\subset X_{n-\alpha} \iff \dim(\xi \cap X_{n-\alpha}) = i - \alpha$ $IC_{\overline{p}}^i(X) = \{ \xi \in C_i(X) \mid \begin{matrix} \dim(\xi \cap X_{n-\alpha}) \leq i - \alpha + p_{\alpha} \\ \dim(\xi \cap X_{n-\alpha}) \leq i - 1 - \alpha + p_{\alpha} \end{matrix} \forall \alpha \geq 2 \}$ $IH_{\overline{p}}^i(X; \mathbb{Z}) = H_i(IC_{\overline{p}}(X))$

pg. 63

e.g.



don't want this one!

★ DEFAULT

⑤. Ex: $X = P^1 \cup P^1 \subseteq P^2$, $X_2 = A^1 \cup A^1 \supseteq X_1 = X_0 = \{pt\}$

$P_0 = P_1 = P_2 = 0$ is only option for perversity fn.

$$IC_i(X) = \left\{ \xi \in C_i(X) \mid \begin{array}{l} \dim(|\xi| \cap X_0) \leq i-2 \\ \dim(|\partial\xi| \cap X_0) \leq i-3 \end{array} \right\}$$

Chains can only touch singularity at $i=2$ (impossible due to this)

$$\Rightarrow IC_*(X) \cong C_*(P^1 \sqcup P^1)$$

$$\text{So } IH_i(X) = H_i(P^1 \sqcup P^1) = \begin{cases} \mathbb{Q} \oplus \mathbb{Q} & i=0,2 \\ 0 & i=1 \end{cases}$$

More generally:

Ex: X is $2m$ -dimensional with an isolated singular point x
 $X = X_{2m} \supset X_0 = \{x\}$.

Let's use the middle perversity

Prop: $IH_i(X, \mathbb{Z}) = \begin{cases} H_i(X, \mathbb{Z}), & i > m \\ \text{Im}(H_m(X-x, \mathbb{Z}) \rightarrow H_m(X, \mathbb{Z})), & i = m \\ H_i(X-x, \mathbb{Z}), & i < m \end{cases}$

Pf: $\xi \in IC_i^m(X)$ iff ① $\dim(|\xi| \cap X_0) \leq i - 2m - \underline{m}(2m) = i - m - 1$
 ② $\dim(|\partial\xi| \cap X_0) \leq i - m - 2$

If $i \leq m$, can't intersect $X_0 = \{x\} \Rightarrow IC_i^m(X) = C_i(X-x)$

If $i \geq m+2$, no condition $\Rightarrow IC_i^m(X) = C_i(X)$

If $i = m+1$, ξ can go through x , but not $\partial\xi$

$$IC_*(X) : \dots \rightarrow C_{m+3}(X) \rightarrow C_{m+2}(X) \rightarrow IC_{m+1}(X) \rightarrow C_m(X-x) \rightarrow \dots$$

$$\quad \quad \quad \hookrightarrow C_{m-1}(X-x) \rightarrow \dots$$

Result is clear for $i \neq m, m+1$.

$$\text{Ker} \left[\underset{\substack{\uparrow \\ C_{m+1}(X)}}{IC_{m+1}(X)} \rightarrow C_m(X-x) \right] = \text{Ker} [C_{m+1}(X) \rightarrow C_m(X-x)]$$

since if $\partial\xi = 0$, $x \in |\partial\xi|$

$\Rightarrow$ result holds at $m+1$

$$\partial : IC_{m+1}(X) \rightarrow \partial C_{m+1}(X) \cap \underbrace{IC_m(X)}_{C_m(X-x)}$$

$\Rightarrow IH_m$ is image of $H_m(X-x)$ in $H_m(X)$

⑥ Cor. $X = \text{pinched torus} = \{y^2 = x^3 - x^2\} \subseteq \mathbb{P}^2$ has

$$H_i(X, \mathbb{Z}) = \begin{cases} \mathbb{Z} & i=0,2 \\ 0 & i=1 \end{cases}$$

Pf: $H_1(\underbrace{X-x}_{\cong S^1}, \mathbb{Z}) \cong \mathbb{Z} \longrightarrow H_1(X, \mathbb{Z})$



$$\begin{array}{ccc} 1 & \xrightarrow{\quad} & 0 \\ & \text{bc gen. is} & \\ & \text{a bdy in } X & \end{array}$$

//

~~Ex: (Why we need a persistence function at all)~~ ← Will return to this later

- Computing with chains is hard; often computed via normalizations.

Normalization: "a type of partial compactification"

Def. A domain A is normal if it is integrally closed in its field of fractions.

ie. $F = \text{Frac}(A) \supseteq A$. A is normal if for every $a_1, \dots, a_m \in A$ & every $x \in F$ s.t. $x^m + a_1 x^{m-1} + \dots + a_m = 0$ we have $x \in A$.

Def. The normalization of A is

$\{x \in F \mid x \text{ satisfies } x^m + a_1 x^{m-1} + \dots + a_m = 0 \text{ for some } m \text{ & } a_i \in A\}$

⑦ Ex: $\{x^2 = y^3\} \subseteq \mathbb{C}^2$ (has singularity at $(0,0)$)

$$\Downarrow$$

$$A = \mathbb{C}[x,y]/(x^2 - y^3) \cong \mathbb{C}[t^3, t^2] \subseteq \mathbb{C}[t]$$

$$\text{Frac}(A) \cong \mathbb{C}(t)$$

t satisfies $t^3 - (x) = 0$ & $\bar{A} = \mathbb{C}[t]$

$$A \longrightarrow \bar{A} \iff \{x^2 = y^3\} \longleftarrow \mathbb{A}^1$$

Normality is related to resolutions of singularities (not equiv. to)

Prop: If A is ^(noetherian) normal, then A is "regular in codim 1"
 i.e. for every height 1 prime $\mathfrak{p} \subseteq A$, $A_{\mathfrak{p}}$ is regular.
 (Geometrically, the singular locus of A is $\text{codim} \geq 2$.)

Pf: Since $A_{\mathfrak{p}}$ is normal, it is sufficient to show?

Claim: If A is a normal, noetherian local ring of dimension 1, then its maximal ideal $\mathfrak{m} \subseteq A$ is principal.

Pf of Claim: $x \in \mathfrak{m}$ nonzero $\Rightarrow \mathfrak{m} = \sqrt{(x)} \Rightarrow \mathfrak{m}^n \subseteq (x)$ some n

Choose n smallest s.t. $\mathfrak{m}^n \subseteq (x)$, so $\exists y \in \mathfrak{m}^{n-1}$ s.t. $y \notin (x)$
 $\Rightarrow t = \frac{y}{x} \in \text{Frac}(A) \setminus A$ & $t \cdot \mathfrak{m} \subseteq A$ so $t\mathfrak{m} = A$ or $\subseteq \mathfrak{m}$.

If $t\mathfrak{m} \subseteq \mathfrak{m} \Rightarrow t\mathfrak{m}$ is f.g. A -module & is $A[t]$ -module

$\Rightarrow t$ is integral $/A \Rightarrow$ contradiction.

So $t\mathfrak{m} = A$ & $\exists m \in \mathfrak{m}$ s.t. $tm = 1 \Rightarrow \mathfrak{m} = A \cdot t^{-1} = A \cdot m$
 is principal //

So the normalization is a "partial resolution".

What does normality look like?

Def. X is topologically normal if $\forall x \in X$, $\exists$ open nbd $U_x \ni x$
 s.t. $U_x - X_{n-2} = U_x - \text{Sing}(X)$ is connected.

Ex:  is not topologically normal, but

 is.

⑦ Prop. X normal $\Rightarrow X$ topologically normal.

Pf: We use a key property of normal varieties:

Zariski's Main Thm: If $f: Z \rightarrow X$ is a proper, birat'l map, X is normal & Z is irreducible, then ① $f^{-1}(x)$ is connected $\forall x \in X$, ② $f_* \mathcal{O}_Z = \mathcal{O}_X$.

Hironaka's Thm: Any quasiprojective variety X admits a (proper) resolution of singularities $f: Z \rightarrow X$.

We argue by passing to $Z: x \in U$ ^{analytic} nhd of x . Then put $E = \pi^{-1}(U) \cap \pi^{-1}(\text{Sing}(X))$. E is clsd in Z & Z is smooth, so $\pi^{-1}(U) \setminus E$ is still connected if $\pi^{-1}(U)$ is connected (in analytic top) (bc $\pi^{-1}(U) = V_1 \cup V_2 \Rightarrow \pi(V_1) \cap \pi(V_2) = \emptyset \Rightarrow U = \pi(V_1) \cup \pi(V_2)$)
 $U \setminus \pi(U)$ — because f is a resolution of singularities //

Thm (Goresky-MacPherson) $\pi: \tilde{X} \rightarrow X$: the normalization of X . Then $\pi_*: IH_i^{\bar{P}}(\tilde{X}; \mathbb{Z}) \rightarrow IH_i^{\bar{P}}(X; \mathbb{Z})$ is an isom.

Pf: π induces a map on complexes $\pi_*: IC_i^{\bar{P}}(\tilde{X}, \mathbb{Z}) \rightarrow IC_i^{\bar{P}}(X, \mathbb{Z})$ which takes a chain $\sigma: \Delta^i \rightarrow \tilde{X}$ to $\pi \sigma$.

If $X_n \supseteq X_{n-1} \supseteq \dots \supseteq X_0$ is strat of X , we can take preims. to get $\tilde{X}_n \supseteq \tilde{X}_{n-1} \supseteq \dots \supseteq \tilde{X}_0$, which suffices to compute $IC_i^{\bar{P}}(\tilde{X})$ (Exercise: $\tilde{X}_n \supseteq \dots \supseteq \tilde{X}_0$ is a Whitney stratification) See Back

We claim any chain $\sigma: \Delta^i \rightarrow X$ lifts uniquely to $\tilde{X}$

Since $\sigma(\text{Interior of } \Delta^i) \subseteq X^{\text{smooth}}$ & π is isom. over X^{smooth} there is unique $\tilde{\sigma}: \Delta^i \rightarrow \tilde{X}$ such that $\pi \tilde{\sigma} = \sigma$

For $a \in \Delta^i$, $B_\epsilon(a) \cap (\Delta^i)^0$ is conn'd, so $\pi^{-1}(B_\epsilon(a) \cap (\Delta^i)^0)$ is connected. $K_\epsilon \subseteq \pi^{-1}(\sigma(\Delta^i))$: cpt (bc π is proper)

$\bigcap_{\epsilon > 0} K_\epsilon$ is non- $\emptyset$ & conn'd (bc they are nested & $\neq \emptyset$ compact sets) $\leadsto$ so $\nexists$ intersection is a pt. & can lift uniquely



not true for arbitrary resolutions of singularities

8.5

Note on functoriality of IH:

Prop. Suppose $f: Y \rightarrow X$ is a finite map. Then there are Whitney stratifications

$$Y = Y_n \supset Y_{n-1} \supset \dots \supset Y_0 \quad \& \quad X = X_n \supset X_{n-1} \supset \dots \supset X_0$$

s.t. $Y_i = f^{-1}(X_i)$.

Pf: Build stratifications together.

At each stage of induction, remove points, their preimages, & their images whenever they fail the 2 Whitney conditions. As this is a closed condition, we can then continue the induction downwards on dimension.

This has applications beyond normal varieties!

e.g. $q: Y \rightarrow Y/G$ for a finite group G acting on a normal variety Y .

Exercise: Show $IH_*^P(Y/G, \mathbb{Q}) = IH_*^P(Y, \mathbb{Q})^G$.

⑨ Ex: Back to Whitney umbrellas:

$$X = \text{Spec}(A) = \{x^2 = y^2 z\} \subseteq \mathbb{C}^3$$

$$A = \mathbb{C}[x, y, z] / (x^2 - y^2 z) \hookrightarrow \mathbb{C}[u, v]$$

$$\begin{array}{ccc} x & \mapsto & uv \\ y & \mapsto & u \\ z & \mapsto & v^2 \end{array}$$

$$A \cong \mathbb{C}[u, uv, v^2] \leftarrow \text{(Can get } v \text{ from } x^2 - (uv)^2)$$

$$\bar{A} = \mathbb{C}[u, v]$$

$$\text{IH}(X; \mathbb{Z}) = \text{IH}(\mathbb{C}^3; \mathbb{Z})$$

For projective varieties, can glue normalizations for affine charts.

e.g. $X = \{x^2 w = y^2 z\} \subseteq \mathbb{P}^3$

On $w \neq 0$: $v_w: \mathbb{A}_{u,v}^2 \rightarrow X$
 $(u, v) \mapsto [uv : u : v^2 : 1]$

On $z \neq 0$: $v_z: \mathbb{A}_{a,b}^2 \rightarrow X$
 $(a, b) \mapsto [a : ab : 1 : b^2]$

glued where $v \neq 0$ / $b \neq 0$ with

$$\left[\frac{u}{v} : \frac{u}{v^2} : 1 : \frac{1}{v^2} \right] = [a : ab : 1 : b^2]$$

$$\boxed{a = \frac{u}{v}}, \boxed{b = \frac{1}{v}}$$

Tot($\mathcal{O}_{\mathbb{P}^1}(1)$) $\xrightarrow{\mathbb{P}^1 \text{ gluing}}$

+ $w = z = 0$ compactifies Tot($\mathcal{O}_{\mathbb{P}^1}(1)$)

$$\Rightarrow \bar{X} = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(1)) \quad \text{"Hirzebruch surface"}$$

$$\text{IH}_i(X, \mathbb{Q}) = H_i(\bar{X}, \mathbb{Q}) \xrightarrow{\uparrow} H_i(\mathbb{P}^1, \mathbb{Q}) \oplus H_{i-2}(\mathbb{P}^1, \mathbb{Q})$$

projective bundle formula
[See back]

$$= \begin{cases} \mathbb{Q}, & i=0 \\ 0, & i=1 \\ \mathbb{Q} \oplus \mathbb{Q}, & i=2 \\ 0, & i=3 \\ \mathbb{Q}, & i=4 \end{cases}$$

Projective Bundle formula: $E \rightarrow X$: vector bundle.

$$H^*(P(E); \mathbb{Z}) \cong \frac{H^*(X; \mathbb{Z})[\xi]}{(\xi^{n+1} + c_1(E)\xi^n + c_2(E)\xi^{n-1} + \dots + c_{n+1}(E))}$$

As $H^*(X; \mathbb{Z})$ - algebras

Cor: $H^k(P(E); \mathbb{Z}) \cong \bigoplus_{i=0}^{\infty} H^{k-2i}(X; \mathbb{Z}) \quad \forall k.$

⑩ Rmk: IH is not a homotopy invariant!

e.g. $S^4 \sqcup \mathbb{C}P^2$

normalization $\searrow$
 $S^4 \cup_{S^2} \mathbb{C}P^2 \sim_{\text{homotopy equiv.}} S^4 \vee S^4$

$S^4 \sqcup S^4$

normalization $\swarrow$

$\Rightarrow IH_i^p(S^4 \vee S^4, \mathbb{Z}) \cong H_i(S^4, \mathbb{Z}) \oplus H_i(S^4, \mathbb{Z}) \leftarrow \text{different!}$
 $IH_i^p(S^4 \cup_{S^2} \mathbb{C}P^2, \mathbb{Z}) \cong H_i(S^4, \mathbb{Z}) \oplus H_i(\mathbb{C}P^2, \mathbb{Z})$

Note: In both formulas above, answer is independent of p ! This reflects smoothness of normalization, as we prove below:

Thm. X : oriented, normal, cpt n -diml pseudomfd / X : projective, normal (real dim n) complex variety.

Then: $IH_i^t(X, \mathbb{Z}) \cong H_i(X, \mathbb{Z})$
 $IH_i^0(X, \mathbb{Z}) \cong H^{n-i}(X, \mathbb{Z})$

$0 = 0$ -perversity fn
 $t = \text{top perversity fn}$
 $t(i) = i - 2, i \geq 2$

[Rmk. Other perversities "interpolate" between H_i & H^{n-i} .]

Pf Ideas: LemA: $IH_0^p(Y) = H_0(Y^{sm})$

[Compute w/ chains: $\dim(\{ \xi \} \cap Y_{n-2}) \leq -2$ for $\xi \in IC_0^p$
 $\& \dim(\{ \xi \} \cap Y_{n-2}) \leq -1$ for $\xi \in IC_1^p$]

$\Rightarrow IH_0^p$ same as computing H_0 for $Y^{sm} = Y - Y_{n-2}$.

LemB L : compact of dim. $k-1$, with Whitney stratⁿ $L = L_{k-1} \supset L_{k-2} \supset \dots \supset L_0$

$IH_i^p(\text{cone}(L)) = \begin{cases} IH_i^p(L) & , i < k-1-p_k \\ 0 & , i \geq k-1-p_k \end{cases}$

Pf: $\text{cone}(L)$ has stratⁿ $\text{cone}(L)_i = \begin{cases} \text{cone}(L_{i-1}) & , i \geq 1 \\ \{v\} & , i = 0 \end{cases}$

$\xi \in IC_i^p(\text{cone}(L))$ is allowable at v iff:

$0 \leq i - k + p_k \iff i \geq k - p_k$

If $i \leq k - p_k - 1$, $\xi \in IC_i^p(\text{cone}(L))$ can't touch v &

$IC_i^p(\text{cone}(L)) \cong IC_i^p(L \times (0,1)) \cong IC_i^p(L) \Rightarrow \begin{cases} \text{for } i < k-1-p_k \\ IH_i^p(\text{cone}(L)) = \\ IH_i^p(L) \end{cases}$

MOVE TO LATER

cone point

⑪ If $i \geq k - p_k - 1$, form $c\xi = \text{cone}(\xi)$ for $\xi \in IC_i^P(\text{cone } L)$

By hypothesis, $\dim(|c\xi| \cap \{v\}) = 0 \leq (i+1) - k + p_k$

$$\& \dim(|c\xi| \cap \text{cone}(L_{k-1-j})) \leq \dim(|\xi| \cap L_{k-1-j}) + 1$$

So $c\xi \in IC_{i+1}^P(\text{cone } L)$. $\leq i-j+p_j+1$

Easy to check: $\partial(c\xi) = \xi \Rightarrow IH_i^P(X) = 0$ for $i \geq k - p_k - 1$

Lem: $IH_*^P(\mathbb{R}^m \times Y) \cong IH_*^P(Y)$

[Pass to normalizations & split as product. Use $H_*(\mathbb{R}) \cong \mathbb{R}$ in deg.]

Mayer-Vietoris for IH: For $U, V \subseteq X$ open, we have a seq of complexes:

$$0 \rightarrow IC_*^P(U \cap V) \xrightarrow{\xi \mapsto (\xi, \xi)} IC_*^P(U) \oplus IC_*^P(V) \xrightarrow{(\alpha, \beta) \mapsto \alpha - \beta} IC_*^P(U \cup V) \rightarrow 0$$

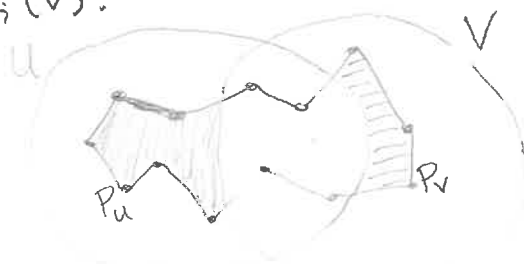
[so we have a les.

$$\dots \rightarrow IH_i^P(U \cap V) \rightarrow IH_i^P(U) \oplus IH_i^P(V) \rightarrow IH_i^P(U \cup V) \xrightarrow{\partial} IH_{i-1}^P(U \cup V)$$

Pf: Most is formal from usual MV sequence.

Hardest Part: If $\xi \in IC_i^P(U \cup V)$, then $\xi = \xi_u + \xi_v$ for $\xi_u \in IC_i^P(U)$ & $\xi_v \in IC_i^P(V)$.

Pf Sketch



define $P = |\xi|$

$f: P \rightarrow \mathbb{R}$ cts. st.

$$f(v) = \begin{cases} 0 & \text{if } v \in \{\text{closed simplex intersecting } P_u\} \\ 1 & \text{if } v \in \{\text{closed simplex intersecting } P_v\} \\ 1/2 & \text{else} \end{cases}$$

For vertices v & extending affinely to P .

Shrink simplices enough s.t. no simplex intersects both P_u & P_v . AND $|\xi| \cap X_i$ are sub-complexes

Pick $c \in (0, 1)$ avoiding vertices &

$\xi_u = \text{part of } \xi \text{ w/ } f < c$ & $\xi_v = \text{part of } \xi \text{ w/ } f > c$.

Compatibility is clear for $|\xi_u|$ & $|\xi_v|$.

$$|\partial \xi_u| \subseteq |\partial \xi| \cup f^{-1}(c)$$

"Thin Wall Lemma": For every subpolyhedron $Y \subseteq P$, $\dim(f^{-1}(c) \cap Y) \leq \dim(Y) - 1$

⑫ Pf of IH^0 & $IH^{\pm}$ for normal X :

Claim: If $F \& G : \{ \text{open sets in } X \} \longrightarrow \{ \text{graded abelian grps} \}$ both satisfy:

- ① $F(U) \longrightarrow F(V)$ for $U \hookrightarrow V$
- ② $F \& G$ satisfy M.V. exact sequences
- ③ $F \& G$ commute w/ direct limits over increasing unions.

Then if $\phi: F \longrightarrow G$ is a natural transformation ^{which} is an isomorphism when $V \cong \mathbb{R}^n \subset X^{sm}$ or $V \cong \mathbb{R}^{n-k} \times \text{cone}(L)$, then ϕ_X is an isomorphism.

Pf: Use MV sequences: • if ϕ is an isom. on $U, V, U \cup V$, then $\phi_{U \cup V}$ is isomorphism.

• Induct on Whitney stratification, covering X_{n-i} at each step

Thm Part 1: $IH_i^{\pm}(X, \mathbb{Z}) \cong H_i(X, \mathbb{Z})$

Pf: need to check for $\text{cone}(L)$:

i	$IH_i^{\pm}(\text{cone } L)$	$H_i(\text{cone } L)$
≥ 1	$\rightarrow 0$	0 [cone L is contractible]
0	$IH_0^{\pm}(L) = H_0(L^{sm})$	$\mathbb{Z}$

$\dim(\mathbb{R}^n \cap X^{sm}) \leq i-2$
 $\forall$ coh degs. k .
 $\Rightarrow IC_*^{\pm} \subseteq C_*^{PL}$
 $\Rightarrow IH_*^{\pm} \rightarrow H_*$

$k-1-(k-2)=1$, so follows from Lem B, then A

Agree when L^{res} is connected $\Leftarrow$ top. normality. //

Thm Part 2: $IH_i^0(X, \mathbb{Z}) \cong H^{n-i}(X; \mathbb{Z})$

$V = \mathbb{R}^{n-k} \times \text{cone}(L)$: map comes from $\sigma \mapsto D(\sigma) =$ dual block of barycentric subdivision.

i	$IH_i^0(V)$	$H_c^{n-i}(V)$
$\leq k-2$	$IH_i^0(L)$	$H^{k-1-i}(L)$
$k-1$	$\rightarrow 0$	$\xleftarrow{\text{when } L \text{ is connected, i.e. normality}} \tilde{H}^0(L)$
$\geq k$	$\rightarrow 0$	0

$k-1-0=k-1$, so Lem B computes

↑ This map requires starting with a triangulation!
 Better: use the cap product. See Back.

Prop. [Cap product factors through IH]: $\forall$ perversity p ,
there is a factorization

$$\begin{array}{ccccc}
 H^{n-k}(X; \mathbb{Z}) & \xrightarrow{\cap [X]} & H_k(X; \mathbb{Z}) & & \\
 \downarrow \text{dashed} & & \uparrow & & \\
 IH_k^p(X; \mathbb{Z}) & \longrightarrow & IH_k^p(X; \mathbb{Z}) & \longrightarrow & IH_k^{\pm}(X; \mathbb{Z})
 \end{array}$$

- ▷ Can be proved using a triangulation of X + explicit formula for cap product.
- ▷ We will defer proof as it follows more cleanly from sheaf-theoretic constructions.

12.5 Borel-Moore Homology & $I^{BM}H$:

(Introduce/motivate BM homology)
 - MV for BM
 - Compute for cones.

X : smooth, but non-compact, oriented mfd.

→ duality & many applications of cohomology require compact supports

ea. $H_c^i(X, \mathbb{Z}) \cong H_{n-i}(X, \mathbb{Z})$

$H^i(X, \mathbb{Z}) \cong \underline{H}_{n-i}^{BM}(X, \mathbb{Z})$

"Borel-Moore" → generalizes well to IH

Borel-Moore Homology: Replaces singular chains $\sum_{k=1}^K \alpha_k \cdot \sigma_k$ with infinite sums $\sum \alpha_\sigma \cdot \sigma$ s.t. $\forall$ compact $K \subseteq X$, only finitely many σ intersect K .

Ex: for $\mathbb{R}$, we have $\sum_{i \in \mathbb{Z}} [i, i+1] \in C_1((\mathbb{R}))$

~~can~~ Can use a triangulation: $\mathbb{R} = \bigcup_{i \in \mathbb{Z}} [i, i+1]$.

$$\begin{array}{ccccc} 0 & \longrightarrow & \prod_{n \in \mathbb{Z}} \mathbb{Z} & \longrightarrow & \prod_{n \in \mathbb{Z}} \mathbb{Z} \longrightarrow 0 \\ & & (a_n) & \longmapsto & (a_{n+1} - a_n) \\ & & H_1 = \mathbb{Z} \cdot (1)_{n \in \mathbb{Z}} & & H_0 = 0 \end{array}$$

Fact: If $\bar{X}$ is cpts of X & $D = \bar{X} - X$,
 $H_c^i(X, \mathbb{Z}) \cong H^i(\bar{X}, D; \mathbb{Z})$

If M is compact, it is clear (from chains) that $H_i = H_i^{BM}$.

Can do the same modification to get $I^{BM}H_i^p$. All results we discussed for IH hold for $I^{BM}H$ (compute by normalizations, MV sequences, etc.).

Goal: $I^{BM}H_i^p(cL) = ?$ for $\dim(L) = k-1$; L : compact (w/o bdy) prebdf.

Sublem: $I^{BM}H_i^p(X; \mathbb{Z}) \cong I^{BM}H_{i+1}^p(X \times \mathbb{R}; \mathbb{Z})$

Pf: Still splits as a product, but $I^{BM}H_i^p(\mathbb{R}, \mathbb{Z}) = \begin{cases} 0 & i=0 \\ \mathbb{Z} & i=1 \end{cases}$

Prop: $I^{BM}H_i^p(cL, \mathbb{Z}) = \begin{cases} IH_{i-1}^p(L, \mathbb{Z}) & i > k-1-p_k \\ 0 & i \leq k-1-p_k \end{cases}$ $\leftarrow$ inequalities reversed from standard case!

Pf: Same condition at cone point: $\dim(\xi \cap \{0\}) \leq i - k + p_k$

so no intersection when $i \leq k-1-p_k$ [when $i - k + p_k \leq -1$]

For $\xi \in IC_i^p(L \times (0, \infty))$ form $R = \sum_{m \geq 0} [m, m+1]$ on $[0, \infty)$ & $\xi \times R$.
 (missing 0)

& consider $\eta = F_*(\xi \times R)$: composing $\xi \times R$ with $F: L \times (0, \infty) \times [0, \infty)^+ \rightarrow L \times (0, \infty)$

~~BM~~ $\partial(\xi \times R) = \partial \xi \times R + (-1)^i \xi \times \partial R \stackrel{\partial \xi = 0}{=} (-1)^{i+1} \xi \times \{0\} \Rightarrow \partial \eta = \xi$

12.75

So this "coring off to infinity" procedure shows

$$I_{H_i}^{\text{BM}}(cL) = 0 \text{ for } i \leq k-1-p_k.$$

For higher degrees $\xi \in IC_i^P(L) \Rightarrow c_\infty(\xi) = \overline{\xi \times R} \approx \xi \times [0, \infty)$

$$c_\infty(\xi) \in I_{i+1}^{\text{BM}}(L)$$

$$c_\infty : IC_*^P(L) \longrightarrow IC_{*+1}^P(cL)$$

$$\partial(c_\infty \sigma) = -c_\infty(\partial \sigma)$$

for $*$ large enough
to go through cone pt. ϵ

//

(13) Poincaré Duality for IH :

Recall: $\pm(i) = \begin{cases} 0, & i \leq 2 \\ i-2, & i > 2 \end{cases}$ is the maximal/top perversity and

$\underline{0}(i) = 0 \forall i$ is the zero perversity.

▷ For p : perversity fcn, define $q = \pm - p$ to be the 'complementary perversity'.

Thm: X : projective complex variety; p & q = complementary.

Then there is a non-degenerate bilinear pairing

$$IH_i^p(X; \mathbb{Q}) \times IH_{n-i}^q(X, \mathbb{Q}) \xrightarrow{\cap} \mathbb{Q}.$$

Notes: ▷ similar versions when X is not projective adding suitable "cpt. supports" conditions.

▷ when X is smooth this is constructed by taking $\sigma_i: \Delta^i \rightarrow X$ & $\sigma_{n-i}: \Delta^{n-i} \rightarrow X$, nudging them until they are transverse, & sending to $\#(\sigma_i! \cap \sigma_{n-i}!)$. Proof of non-degeneracy comes from local computations + M.V. sequence

Idea for proof is similar: take chains s.t. $\sigma_i! \cap \sigma_{n-i}! \subseteq X^{\text{sm}}$ & is finitely many points. Using M.V., reduces to showing the corresponding statement for cones $\text{cone}(L)$.

Recall: for $\dim(L) = m$,

$$IH_i^p(\text{cone } L, \mathbb{Z}) = \begin{cases} IH_i^p(L; \mathbb{Z}) & , i < m - p_{m+1} \\ 0 & , \text{else} \end{cases}$$

$$I^{BM}H_i^p(\text{cone } L, \mathbb{Z}) = \begin{cases} 0 & , i \leq m - p_{m+1} \\ IH_{i-1}^p(L; \mathbb{Z}) & , \text{else} \end{cases}$$

If L satisfies P.D. (induction on $\dim$), then $\forall i < m - p_{m+1}$

$$IH_i^p(\text{cone } L, \mathbb{Q}) \cong IH_i^p(L, \mathbb{Q}) \xrightarrow{\downarrow} IH_{n-i}^q(L, \mathbb{Q})^* \cong I^{BM}H_{m+1-i}^q(\text{cone } L, \mathbb{Q})^*$$

& if $i \geq m - p_{m+1}$, since $m-1 = p_{m+1} - q_{m+1}$,

$$IH_i^p(\text{cone } L, \mathbb{Q}) = 0 = I^{BM}H_{m+1-i}^q(\text{cone } L, \mathbb{Q})$$

Later, we'll return. & reprove this with sheaves/Verdier duality.

Problem 1. Find the point on $y = -2x - 1$ closest to $(1, 2)$.

Problem 2. Does Mean Value Theorem apply for

$$f(x) = \begin{cases} x^2 + x, & 0 \leq x < 1 \\ 3x - 1, & 1 \leq x \leq 2 \end{cases}$$

on the interval $[0, 2]$?

Problem 3. Approximate $\sqrt{8.99}$ (without a calculator).

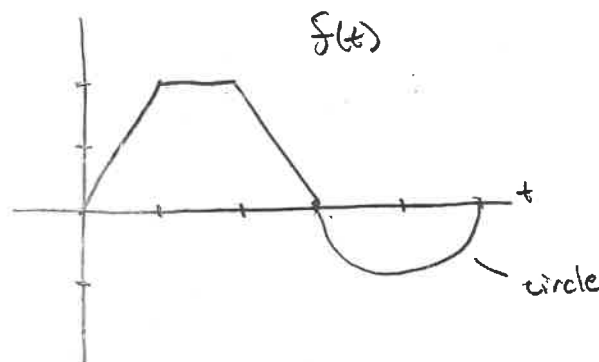
Problem 4. Find the definite integral $\int_0^3 2x^2 - x \, dx$ using limits of Riemann sums.

Problem 5. 1. When is R_n an overestimate for the area under a function? How about L_n ?

2. Approximate $\int_1^3 3x^2 + x \, dx$ using left approximation with 4 rectangles. Without computing the true value of the integral/area, is it an over or underestimate?

Problem 6. A cylindrical can is to be made to hold 250 mL of oil. Find the dimensions that minimize the cost of material to make the can.

Problem 7. (Leftover from Monday's worksheet) If f is a function with graph below and $g(x) = \int_0^x f(t) \, dt$, then find $g(0), g(1), g(2), g(3), g(4), g(5)$. Sketch the graph of $g(x)$.



(13.5) Remark: Here, the complementary perversities appear in local calculations (for cones). Another way to see complementary perversity fns must appear is through the heuristics of intersection #'s:

~~an~~ an allowable i -chain σ_i & $n-i$ chain σ_{n-i} must not meet on X_{n-k} . Their expected intersection dim. is

$$(i-k+p_k) + (n-i-k+q_k) - (n-k) = p_k + q_k - k.$$

To make this < 0 , need $p_k + q_k \leq k-1$.

For homologies to also miss each other: $p_k + q_k \leq k-2$

Making this equal allows "as many cycles in dual as possible" $\longrightarrow$ makes pairing non-degenerate.

(14) Note there are 2 "middle" perversities:

$$m = [0, 0, 0, 1, 1, 2, 2, 3, 3, \dots] \text{ "upper middle"}$$

$$n = [0, 0, 0, 0, 1, 1, 2, 2, 3, 3, \dots] \text{ "lower middle"}$$

$m+n = t$ are complementary.

For general pseudomanifolds (locally conical mfd's), $IH^m \neq IH^n$
e.g. real algebraic varieties don't necessarily have $IH^m = IH^n$

Prop. If all strata in a Whitney stratification of X are even dimensional, then $IH_*^m(X) = IH_*^n(X)$.

ex: X : complex algebraic variety.

Pf: Since strata are even-dim'l, they only see $p(2i)$ for $\dim_{\mathbb{C}}(X) = \frac{1}{2} \dim_{\mathbb{R}}(X)$. //

In this case, $IH_*^m(X) = IH_*(X)$ is self-dual under Poincaré Duality.

e.g. If X : cx alg. variety has smooth normalization, then $IH_*^p(X) = IH_*(X)$ $\forall$ perversity fcn's p & so $IH_*^p(X)$ satisfy Poincaré Duality.

Introduces new invariants associated to X

e.g. If X has even complex dimension $2n$, get a non-deg. pairing $IH_{2n}^m(X, \mathbb{Q}) \otimes IH_{2n}^n(X, \mathbb{Q}) \rightarrow \mathbb{Q}$.

$$\sigma(X) := \#(\text{positive eigvals}) - \#(\text{negative eigvals})$$

"signature of X " $\rightarrow$ topological (bordism) invariant of X .

①5 Crash Course on Sheaf Theory:

New Goal: Sheafify everything we talked about.

$I\mathcal{H}_i \rightsquigarrow IC$ sheaves

We want this for a few reasons:

- ▷ Can start to deal with families.
- ▷ Can understand relationships between IC sheaves on subvarieties of ambient spaces/sources & targets of maps
- ▷ Categorifies constructions in geometric rep. theory, number theory/Langlands program, etc.

Def. A presheaf is a contravariant functor

$$F: \left\{ \begin{array}{c} \text{open sets} \\ \text{in } X \end{array} \right\} \longrightarrow \left\{ A\text{-modules} \right\}$$

$$F(U) =: H^0(U, F) =: \Gamma(U, F) \quad \left(\text{e.g. abelian grps.; } \mathcal{O}_X\text{-modules; vector spaces; ...} \right)$$

F is a sheaf if it also satisfies the gluing property:

• If $s_i \in \Gamma(V_i, F)$ s.t. $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j} \forall i, j$

Then $\exists! s \in \Gamma(\bigcup_i V_i, F)$ s.t. $s|_{V_i} = s_i \forall i$

[Note: might be infinitely many V_i !]

e.g. ① $M \in A\text{-mod}$.

$\underline{M}_X =$ constant sheaf on X

$$\underline{M}_X(U) := \left\{ \text{cts. } f: U \rightarrow M \right\} \cong M^{\# \pi_0(U)}$$

② "Structure sheaf" of X , $\mathcal{O}_X = \{ \text{sheaf of fns on } X \}$

If X is a scheme & $U = \text{Spec}(R) \subset X$ is an affine patch, $\mathcal{O}_X(U) = R$.

Def. The stalk of F @ $x \in X$ is $F_x = \varinjlim_{x \in U} F(U)$

①6

There are 2 flavors of sheaves that will be interested in: coherent & constructible

Coherent: Any algebraic variety, X , comes with a structure sheaf $\mathcal{O}_X$ of fns on X .

$$\mathcal{O}_X(U) = \{ \text{algebraic fns on } U \}$$

A coherent sheaf is a sheaf $\mathcal{F}$ on X s.t.

$$\mathcal{F}(U) \text{ is a fin. presd } \mathcal{O}_X(U)\text{-module for all small enough } U$$

If $U = \text{Spec}(A) \subseteq X$ is an affine (Zariski) open set,

$$\mathcal{F}|_U = \tilde{M} \text{ for a fin. } A\text{-module } M$$

(i.e. $\mathcal{F}(U - \{a\}) = M_{(a)}$)

Constructible:

Def. An A -local system on X is a locally constant sheaf $\mathcal{L}$ of A -mods. i.e. $\exists$ open covering $X = \bigcup U_i$ s.t. $\mathcal{L}|_{U_i} \cong M_i$

for an A -mod M_i .

$\triangleright$ If X is connected $M_i = M \forall i$ is constant.
 $\triangleright$ If $M = M_i = A \forall i$, then $\mathcal{L}$ is of "rank n ".

Exs: ① (from topology) $\mathcal{O}_X^{\text{naive}}(U) = H_n(X, X - U; A)$
 $\mathcal{O}_X^{\text{naive}} = (\mathcal{O}_X^{\text{naive}})^{\text{sheaf}}$

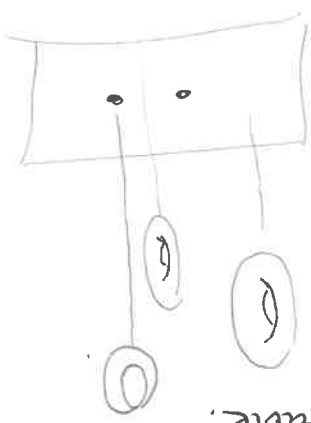
$\mathcal{O}_X$ is constant $\Leftrightarrow X$ is orientable.

② (family) $X = \{y^2 = x(x-1)(x-\lambda)\}$



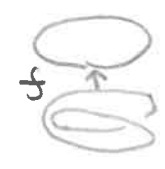
$A_1^2 \setminus \{0, 1\}$

$F(U) = H_1(f^{-1}(U), \mathbb{Z})$ stalls are $\mathbb{Z}^2$



How to tell what a locally constant sheaf is? (Or even whether it is nontrivial??)

A: Walk along paths in X & see if you end up in the same place!



Ex: Consider double cover of S^1
 $f(u) := f^* \Delta_{S^1}(u) = \Delta_{S^1}(f^{-1}(u))$

$f^* \cong A^2$ (rank 2 local system) is locally constant.

(Can trivialize $S^1 = U_1 \cup U_2$)

$$f|_{U_i} \cong A^2_{U_i}$$

$$(a_1, a_2) \in f(U_1) \rightarrow (a_1, a_2) \in f(U_2) \rightarrow (a_2, a_1)$$

$$\rightarrow \pi_1(S^1, x_0) \rightarrow \text{Aut}(A^2) \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

order 2 image

Prop. X : connected top. space. Then TFAE:
 (a) A - local systems on X w/ fibers M
 (b) representations $\rho: \pi_1(X, x_0) \rightarrow \text{Aut}(M)$.

Pf: ① ρ : local system $\Rightarrow \rho: \pi_1(X, x_0) \rightarrow \text{Aut}(M)$

Bc: $\rho: [0, 1] \rightarrow X$ gives a trivial local system
 $\rho^{-1} \rho_0 \cong \rho_0$ & $\rho^{-1} \rho_1 \cong \rho_1$.

② $\rho: \pi_1(X, x_0) \rightarrow \text{Aut}(M) \Rightarrow \rho$: local system

Bc: ρ = sheaf of sections of $(\tilde{X} \times M)/\pi_1(X, x_0)$

for $\tilde{X} \rightarrow X$: universal cover.
 Sheaf of sections of $(\tilde{X} \times M)/\pi_1(X, x_0)$ has monodromy:
 $\rho_p = \rho$

$T_x \pi_1(X, x_0)$ acts by $T_x(s(\tilde{g} \cdot x_0)) = s(\tilde{g}(1)) = s(\tilde{g}(1)) = \rho(\tilde{g})s(x_0)$
 $s \circ \rho^{-1}(u) \rightarrow M$ loc. const. s.t. $s(g \cdot \tilde{x}) = \rho(g) \cdot s(\tilde{x})$

$$\tilde{\rho} = \tilde{\rho}: \tau: \tilde{M} \xrightarrow{\sim} p^* \mathcal{L} \text{ Deck gp. } \pi_1, \tau_{g\tilde{x}} = T_{\alpha} \circ T_{\tilde{x}} = \tau_{\tilde{x}} \circ \rho(\tilde{g})$$

18 (Harder example) Recall the elliptic curve example.

Consider $\mathcal{E}(u) = H_1(f^{-1}(u))$ fibers $\mathbb{Z}^2$ $\pi_1(A_1^2 - \{0, 1\}) \cong F_2$.

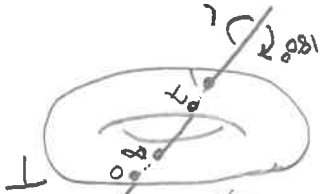
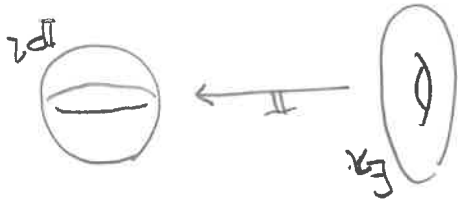
$\mathcal{Q}$: What is $p: F_2 \rightarrow \text{Aut}(\mathbb{Z}^2)$ corresponding to $\bar{A}$:

$\bar{A}: p(e_i) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ "Dehn twist"



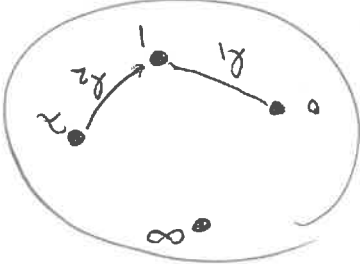
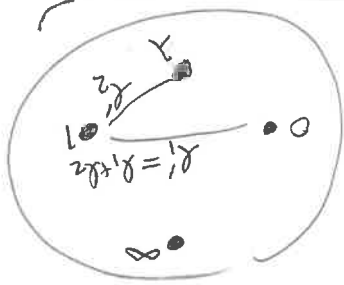
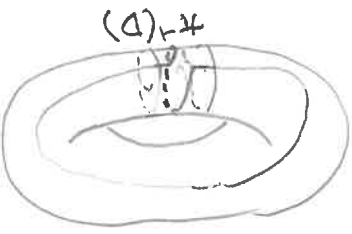
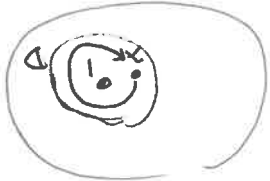
(here in Girard's notes...)

$\overline{pf}$:



Topologically: π_2

$T/\mathbb{Z} \cong S^2$



In larger family $X = \{y^2 = (x+a)(x-b)(x-c)\}$

Pf. of Ehresmann's Thm: $f: X \rightarrow Y$: ^{proper} submersion of smooth ^{mfd} mfd
 Ehresmann is local on target: suffices to ~~try~~ prove for $Y = \mathbb{R}^n$
 Submersions have local coordinates @ $x \in X$ given by
 $\mathbb{R}^{n+m} \xrightarrow{\text{proj}} \mathbb{R}^n$

Choose partition of unity for open cover of $X \Rightarrow$ lift $\partial_i \in \mathcal{X}(\mathbb{R}^n)$
 to $v_1, \dots, v_n \in \mathcal{X}(X)$ $[v_i = \sum_{\alpha} p_{\alpha} \frac{\partial}{\partial x_i^{\alpha}}]$ s.t. $df_p(v_i) = \partial_i \forall p$.

$\varphi^i =$ flow along v_i ; $\Phi: \mathbb{R}^n \times f^{-1}(d) \rightarrow X$ by
 $\Phi(s_1, \dots, s_n, p) = \varphi_{s_n}^n \circ \dots \circ \varphi_{s_1}^1(p)$ is an isom.

[Uses properness to show flow is well-defined/doesn't blow up & easy to check $f(\Phi(s, p)) = s$.]

Lem: $H^1(X, \mathcal{L}) \cong M_{\pi_1(X, x_0)}^{\text{fiber of } \mathcal{L} @ x_0}$ when X is conn'd

Pf: $H_0(X, \mathcal{L}) \cong M_{\pi_1(X, x_0)} = M / \langle \delta_{*} m - m \rangle$

Pf: $H_0(X, \mathcal{L}) = \left(\bigoplus_{x \in X} \mathcal{L}_x \right) / \left\langle (\alpha_* m)_{\alpha(1)} - m_{\alpha(0)} \mid \alpha = \text{path } \delta m \in \mathcal{L}_{\alpha(0)} \right\rangle$
 (1-simplex)

$\left[\phi(m_x) = u_{x,*}^{-1}(m_x) \right]$ use path to transport to,
 $M_{\pi_1(X, x_0)}$

$\triangleright \phi$ is well-defined bc. $\gamma' = u_x \circ u_x^{-1}$ is loop
 so $\gamma_* \circ u_{x,*}^{-1}(m_x) - u_{x,*}^{-1}(m_x) = 0$.
 $\parallel$
 $u_{x,*}^{-1}(m_x)$

$\triangleright$ Relations exactly match after ϕ . //

Claim: $f: X \rightarrow Y$: smooth + proper $\Rightarrow f_* \mathcal{L}$ is a local system if $\mathcal{L}$ is.

2 Main Ingredients:

① Proper base change: Suppose we have a Cartesian diagram:

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & \lrcorner & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

If f is proper & F is constructible, then $g^* f_* F \cong f'_* g^* F$.

Properness used for openness of fibers & "tube lemma" $\lim_{\leftarrow} F(f^{-1}V) \cong \Gamma(X_s, F|_{X_s})$

Cor:

$$\begin{array}{ccc} X & \rightarrow & X \\ \downarrow f & & \downarrow f \\ \text{Spec}(k) & \rightarrow_S & S \end{array}$$

Then $(F|_{X_s})(X_s) = f_* F|_s$ for $s \in S$.

i.e. fiber of $f_* F$ is $\Gamma(F|_{\text{fiber}})$.

Notes: proper is necessary! $f: \mathbb{A}^2 \setminus \{(0,0)\} \xrightarrow{\pi_2} \mathbb{A}^1$
 $f_* \mathcal{O}_{\mathbb{A}^2 \setminus \{(0,0)\}} \cong \mathcal{O}_{\mathbb{A}^1}[x]$ & $\Gamma(\mathcal{O}_{\mathbb{A}^1 \setminus \{0\}}, \mathbb{A}^1 \setminus \{0\}) = k[x, x^{-1}]$
 $\Rightarrow f_* \mathcal{O}|_0 = k[x] \leftarrow \text{not equal!}$

MORE ON BACK!

② Ehresmann's Thm: A proper submersion of smooth mfd's is a locally trivial fibration, i.e. $\forall y \in Y, \exists \text{ neighd } U \ni y \text{ s.t. } f^{-1}(U) \cong U \times f^{-1}(y)$.

~~By ②, $f^{-1}(U) \cong U \times f^{-1}(y)$ is locally trivial & $\mathcal{L}$ is locally trivial.~~

Δ Can argue claim only with ② actually! ① is just needed to say $f_* \mathcal{L}$ measures $H^i(X_y, \mathcal{L}|_{X_y})$.

Prop: $f: X \rightarrow Y$: proper map & $\mathcal{L}$: local system (or constructible sheaf) on X .

Then $f_* \mathcal{L}$ is constructible on Y .

Pf: uses a powerful result on Whitney stratifns:

* Thom's first isotopy lemma: If $X = \bigsqcup_{\lambda} X_{\lambda}$ is Whitney stratⁿ s.t. $X_{\lambda} \rightarrow Y$ is smooth, then $f: X \rightarrow Y$ is a stratified-locally-trivial fibration. } have non-smooth X

* Existence: Build a Whitney stratⁿ of X & Y s.t.:

- $\Delta f^{-1}(Y_{\lambda})$ is a union of strata of X
- $\Delta f_{\lambda}: X_{\lambda} \rightarrow Y_{\lambda}$ is a submersion/smooth when it is defined.
- $\Delta (\bigsqcup_{\lambda} X_{\lambda})$ has $\mathcal{L}|_{X_{\lambda}}$ locally constant.

* Then $f_* \mathcal{L}|_{Y_{\lambda}}$ is a local system by isotopy lemma $\Rightarrow f_* \mathcal{L}$ is constructible //

(Co)Homology with coefficients in a local system:

for $\sigma: \Delta^i \rightarrow X$ & $\mathcal{L}$: local system $\Rightarrow \sigma^* \mathcal{L}$: loc. sys. on Δ^i
 Note Δ^i = contractible, so $\sigma^* \mathcal{L}$ = constant shf.

$\mathcal{L}_\sigma := \sigma^* \mathcal{L}(\Delta^i)$ = fiber of $\mathcal{L}|_{\sigma(\Delta^i)}$; $\forall p \in \Delta^i$, get restⁿ map
 $p \in \tau \subset \sigma: P_\tau^\sigma = (P_\tau^\tau)^{-1} \circ P_p^\sigma: \mathcal{L}_\sigma \rightarrow \mathcal{L}_\tau$ $P_p^\sigma: \mathcal{L}_\sigma \xrightarrow{\sim} \mathcal{L}_{\sigma(p)} \cong (\sigma^* \mathcal{L})_p$
 $C_i(X, \mathcal{L}) = \left\{ \sum_\sigma \mathcal{L}_\sigma \cdot \sigma \mid \Sigma \text{ is finite (modify for BM)} \right\}$

$$\partial_i \left(\sum_\sigma \mathcal{L}_\sigma \cdot \sigma \right) = \sum_\sigma \sum_{\tau: \text{face of } \sigma} \pm P_\tau^\sigma(\mathcal{L}_\sigma) \cdot \tau$$

only difference from normal homology.

$$H_i(X, \mathcal{L}) = \text{Ker}(\partial_i) / \text{Im}(\partial_{i+1})$$

Ex: $H_i(\mathbb{C}^x, \mathcal{L}_m) = ?$ [$\mathcal{L}_m$ = local system for monodromy $z \mapsto z^m$]
 e.g. $\mathbb{Z} \rightarrow \text{Aut}(A) = A^x$
 $n \mapsto \lambda^n$

$\mathbb{C}^x \sim S^1$, so suffices to do S^1 : $S^1 = \bigcirc$ has triangulation T w/ $\sigma: \Delta^1 \rightarrow S^1$

$$C_1^T(\mathbb{C}^x, \mathcal{L}_m) = C_0^T(\mathbb{C}^x, \mathcal{L}_m) = A.$$

$$0 \rightarrow A \xrightarrow{\lambda - \text{Id}_A} A \rightarrow 0$$

the isomorphism matters!

$$H_d(\mathbb{C}^x, \mathcal{L}_m) = \text{Ker}(\partial_{m-1}) = \begin{cases} 0 & \text{if } m \neq 1 \\ k & \text{if } m = 1 \end{cases}$$

$$H_1(\mathbb{C}^x, \mathcal{L}_m) = k / (\partial_{m-1}) = \begin{cases} 0 & \text{if } m \neq 1 \\ k & \text{if } m = 1 \end{cases}$$

Can define similarly for $\mathbb{I}H_i$

Have all similar formulas as H_i : MV seqs, [Poincaré duality], if fibers are vs.: $\mathbb{I}H_i(X, \mathcal{L}) \cong \mathbb{I}H^i(X, \mathcal{L})^*$

$\Delta C_*(X, \mathcal{L}) = C_*(X_H) \otimes_{G/H} M$ & $C^*(X, \mathcal{L}) = \text{Hom}_{G/H}^*(C_*(X_H), M)$.
 $\text{HCKer}(P_2) \xrightarrow{\uparrow} \text{fiber}$ $\Delta H^0(X; \mathcal{L}) \cong M^{\pi_1(X)}$

①9 local systems are clsd under $\oplus, \otimes, \underline{\text{Hom}}, (-)^V = \underline{\text{Hom}}(\mathbb{Z}^V, \mathbb{A}_X)$

In fact, they are even better:

Prop: $\text{Loc}(X) \subset \text{Sh}(X)$ is a full abelian subcat.

Pf: $\mathcal{L} \xrightarrow{\phi} \mathcal{L}'$ a map of local systems.

$$\text{Ker}(\phi)(U) := \text{Ker}(\mathcal{L}(U) \xrightarrow{\phi(U)} \mathcal{L}'(U)) \leftarrow \text{no sheafification}$$

locally constant.

$$\text{Coker}(\phi)(U) := \text{Coker}[\mathcal{L}(U) \rightarrow \mathcal{L}'(U)]^{\text{Sh}}$$

Sheafification

Note: f_* is same as coherent case, but f^* is not!
 $f^*F = f^{-1}F$

Note: NOT true for vector bundles for example:

On $\mathbb{P}^1$, have $0 \rightarrow \mathcal{O}(-1) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_p \rightarrow 0$.

$f^{-1}F|_V = M|_V$ for $V = f^{-1}(U)$ s.t. $F|_U \neq 0$

Prop. $\text{Loc}(X)$ is stable under f^* & f_* for finite covering maps.

WARNING: Not stable under f_* for general proper maps!

(fibers can jump for example!)

Constructible Sheaves: These will capture $Rf_* \mathbb{K}_X \forall f: X \rightarrow Y$ (proper)

Def. $F \in (\mathbb{K}_X\text{-mod})$ is constructible if $X = \coprod_{\lambda} X_{\lambda}$ is stratified so that $F_{X_{\lambda}}$ is a local system $\forall \lambda$.

Ex: $f: \mathbb{C} \rightarrow \mathbb{C}$ by $z \mapsto z^m$

$f_* \mathbb{K}_{\mathbb{C}} = ?$ For $D = \text{disk not thru } 0$,

$$f_* \mathbb{K}_{\mathbb{C}}(D) = \mathbb{K}_f(f^{-1}(D)) \cong \mathbb{K}^m$$

For $D = \text{disk thru } 0$, $f_* \mathbb{K}_{\mathbb{C}}(D) \cong \mathbb{K}$.

$$f_* \mathbb{K}_{\mathbb{C}}|_{\mathbb{C}^*} \cong \mathcal{L}_{(z \mapsto z^m \text{ on } \mathbb{C}^*)} \quad \& \quad f_* \mathbb{K}_{\mathbb{C}}|_0 \cong \mathbb{K}$$



$$\mathbb{K} = f_* \mathbb{K}_{\mathbb{C}}(D) \xrightarrow{\text{res.}} f_* \mathbb{K}_{\mathbb{C}}(D') = \mathbb{K}^{\oplus m}$$

is invts. under monodromy rep.

(20)

This example is more general than it seems!

If $f: C \rightarrow C'$ is a non-constant map b/w smooth curves, locally it is just $z \mapsto z^{e_x}$ for $e_x = \text{ramification index @ } x$.

[Ramification pts. are $x \in C'$ where f is not smooth!]

Detour: smooth maps in algebraic geometry.

"smoothness of a map $f: X \rightarrow Y$ " $\iff$ "submersion of (pseudo)manifolds"

Def: $\text{Spec}(A[x_1, \dots, x_n]/(g_1, \dots, g_r)) \rightarrow \text{Spec}(A)$ is smooth if $\left(\frac{\partial g_i}{\partial x_j}\right)_{\substack{1 \leq i \leq r \\ 1 \leq j \leq n}}$ is of rank r $\forall x$.

i.e. $f: X \rightarrow Y$ is smooth iff $\nRightarrow$ it induces surjections on tangent spaces.

non-ex: $\pi: \text{Spec } k[u, v] \rightarrow \text{Spec } k[x, y]$
by $x \mapsto u, y \mapsto uv$.

$$k[u, v] \cong k[x, y][v]/(xv - y)$$

$[x]$ vanishes @ $x=0$.

Thm: $\varphi: X \rightarrow Y$ is smooth iff φ is flat, locally of finite presentation and its fibers are smooth.

E.g. • $\varphi: \mathbb{A}^1 \rightarrow \mathbb{A}^1$ by $z \mapsto z^m$

Fiber @ 0 is $k[x]/(x^m) \leftarrow \text{non-reduced, so not smooth!}$

- Elliptic curve ex. is smooth on $\mathbb{A}^1 - \{0, 1\}$ but not if we extend to 0 or 1

Recap on abelian categories:

Working Def. An abelian category $\mathcal{A}$ is an additive category
(i.e. s.t. $\text{Hom}_{\mathcal{A}}(A, B)$ is an abelian gp + mild conditions.) s.t.

$\forall A \xrightarrow{\phi} B$ in $\mathcal{A}$, $\exists \text{Ker}(\phi), \text{Coker}(\phi) \in \mathcal{A}$ &

$\text{Coim}(\phi) \rightarrow \text{Im}(\phi)$ is an isomorphism.
(quotient of A) (sub of B)

Defined via universal property: e.g. $\text{Coker}(\phi) \in \mathcal{A}$ is

$$A \xrightarrow{\phi} B \longrightarrow \text{Coker}(\phi)$$

$$\begin{array}{ccc} & \searrow \omega & \downarrow \exists! \\ \forall f & \searrow & \downarrow \\ \text{s.t. } f \circ \phi = 0 & & \end{array}$$

Exs: $\triangleright \text{Mod}_A = \text{cat. of } A\text{-modules}$
 $\triangleright \text{Sh}(X) = \text{cat. of sheaves on } X$

This condition arises when one is not careful w/ decorations
one adds: e.g. $\{\text{filtered vector spaces}/k\}$ is not abelian, but has
Kernels & Cokernels.

Problem: $F^p k := \begin{cases} k & \text{for } p \geq 1 \\ 0 & \text{for } p < 1 \end{cases}$ & $\tilde{F}^p k = \begin{cases} k & \text{for } p \geq 0 \\ 0 & \text{for } p < 0 \end{cases}$

$f: k \rightarrow k$ is filtered, i.e. $f(F^p k) \subseteq F^p k$.

$\text{coim}(f) = k$ w/ F -filtⁿ

$\text{im}(f) = k$ w/ $\tilde{F}$ -filtⁿ

(21)

Sources of Constructible Sheaves: Cohomology of proper maps!

Next Goal: develop some homological alg. to define $R^i f_* \mathcal{L}; H^i(X, \mathcal{L})$.

Reminders on sheaf cohomology:

- Example of a derived functor: measures discrepancy of a functor from being exact.

between abelian cats

Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a functor. F is right exact (resp. left exact) if whenever $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$ is s.e.s., we have an exact seq $F(A_1) \rightarrow F(A_2) \rightarrow F(A_3) \rightarrow 0$ (resp. $0 \rightarrow F(A_1) \rightarrow F(A_2) \rightarrow F(A_3)$)

Exs: f_* is left exact; $\Gamma(X, -)$ is left exact
 $- \otimes F$ is right exact.

Thm/Def: $F: \mathcal{A} \rightarrow \mathcal{B}$ a left exact functor. Then its right derived functors $R^i F: \mathcal{A} \rightarrow \mathcal{B}$ are functors s.t.:

- $\triangleright R^i F$ is additive & $= 0$ for $i < 0$.
- $\triangleright R^0 F = F$
- $\triangleright$ Have a long $\forall$ ses.
- $\triangleright$ functorial as map on ses
- $\triangleright R^i F$ is universal with these properties
- $\triangleright \forall A \in \mathcal{A}, A \hookrightarrow I$ s.t. $R^i F(I) = 0 \forall i > 0$

Some examples: $\triangleright \Gamma(X, -)$ has right derived functor $H^i(X, -)$

$\triangleright$ Derived functors of $f_*(-)$ gives $R^i f_*(-)$: sheaves whose fibers are coh. of fibers (proper base change!)

★ Thm/Def does not show how to compute! ★

Practical Guide: ~~replace~~ To compute $F(A)$, replace A by a complex $A \rightarrow C^\bullet$.

For all of our purposes, we want A & $C^\bullet$ to be "the same"

$\triangleright$ In particular, need $H^i(A) = H^i(C^\bullet)$, but this is not enough!

$\triangleright$ Rather than give formulas, let's first explore what we mean by 2 complexes being "the same".

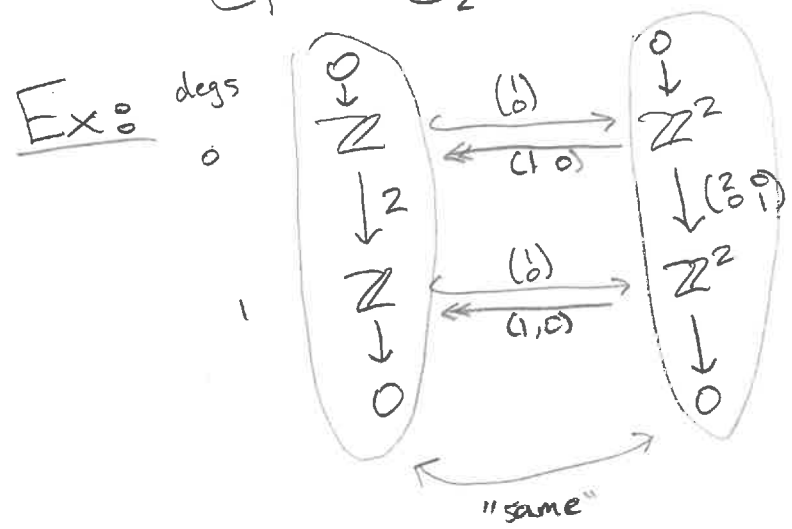
The Derived Category:

~~Abelian~~ A : abelian category (e.g. $\text{Loc}(X)$, $\text{Sh}(X)$, etc.)

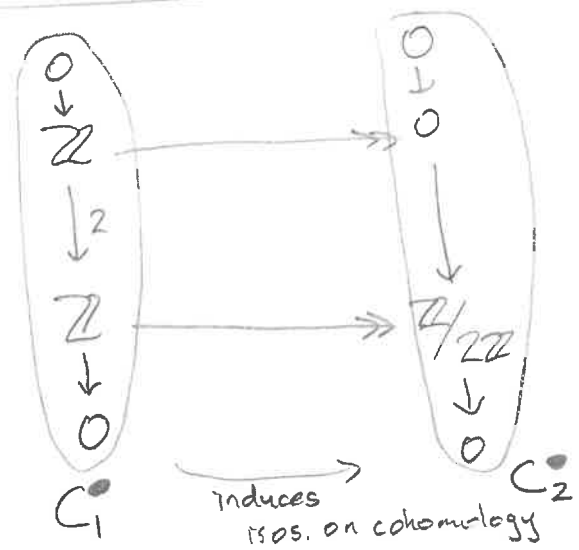
$C(A)$ = category of complexes

$$\dots \rightarrow A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow \dots \text{ in } A$$

IDEA: want to treat 2 complexes as "the same" if $C_1 \rightarrow C_2$ induces isomorphisms on (co)homology groups



call these "quasi-isomorphisms"



But: there's no inverse!

Working Def: $D(A)$ is a category whose objects are complexes & maps are usual maps + inverses of all quasi-isos.
e.g. $C_1 \cong C_2$ in $D(A)$.

Cost of this construction is that $D(A)$ is no longer abelian!

Rmk: $R^i f_* \mathcal{L}$ is constructible for $\mathcal{L}$ constructible on X
 & $f: X \rightarrow Y$ proper. (Uses same proof as before!)

→ Def: $D_c(X)$ [or $D_c^b(X)$] denotes the full sub-cat.
 of $D(X) = D(\text{Sh}(X))$ [or $D_c^b(X) = D^b(\text{Sh}(X))$] of $C^\bullet$
 s.t. $H^i(C^\bullet)$ is constructible $\forall i$.

assume
also all
local systems
involved
are finite
rank!

★ $\text{Constr}(X) :=$ category of constructible shvs on X .

Prop: $D^b(X) \cong D^b(\text{Constr}(X))$

Pf Idea: for $\mathcal{L} \in \text{Constr}(X)$ & $F \in \text{Sh}(X)$, s.t. $F \twoheadrightarrow \mathcal{L}$,
 $\exists \mathcal{L}' \rightarrow F$ s.t. $\mathcal{L}' \rightarrow F \rightarrow \mathcal{L}$ is surj. & $\mathcal{L}' \in \text{Constr}(X)$
 Build $\mathcal{L}'$ by lifting local gens of $\mathcal{L}$ to F

Philosophy: complexes are better than cohomology.

Exs: ① $R(F \circ G) \cong RF \circ RG$ for left exact F, G

[vs. spectral sequence $R^p F(R^q G) \Rightarrow R^{p+q}(F \circ G)$]

② Universal coeffs: $C^\bullet =$ cx computing $H^i(X, \mathbb{Z})$.

cx computing $H^i(X; A)$ is $C^\bullet \otimes_{\mathbb{Z}}^L A$

(vs. $H^i(X; A) \cong (H^i(X, \mathbb{Z}) \otimes_{\mathbb{Z}} A) \oplus \text{Tor}^1(H^{i+1}(X, \mathbb{Z}), A)$)

Consequence: Often later try to work directly in $D_c^b(X)$.

Next Goals: ▷ 6 functor formalism in $D_c^b(X)$ [& introduction to $f_!, f^!$]

▷ Verdier duality in $D_c^b(X)$

→ opens up calculus of the derived category!

f_* & f^* are adjoint functors:

Prop. $f: X \rightarrow Y$. This is a canonical isom. (functorial in each input)

$$\text{Hom}(f^*F, G) \cong \text{Hom}(F, f_*G)$$

Rmk. In coherent category, this is locally:

$$\begin{array}{ccc} X & \xrightarrow{f^*} & Y \\ \downarrow & & \downarrow \\ B & \xrightarrow{f^*} & C \end{array} \quad \text{Hom}_B(M \otimes_A B, N) \cong \text{Hom}_A(M, N)$$

Special Case of:

$$\text{Hom}_B(M \otimes_A P, N) \cong \text{Hom}_A(M, \text{Hom}_B(P, N))$$

Pf. $\phi \in \text{RHS}$, $F \rightarrow f_*G$.

$$\forall V \supset f^*(U), \quad f^*(V) \supset U$$

$$\Rightarrow F(V) \rightarrow f_*G(V) = G(f^*V) \xrightarrow{\text{res}} G(U)$$

$$\text{gives } \varinjlim_{V \supset f^*(U)} F(V) \rightarrow G(U).$$

In reverse: $\psi: f^*F \rightarrow G$.

$\forall V \subset Y$ open, $f(f^*V) \subset V$ gives

$$F(V) \rightarrow \varinjlim_{W \supset f(f^*(V))} F(W) = f^*F(f^*(V)) \rightarrow G(f^*(V))$$

Can check these are inverse operations on stalks //

$$\parallel f_*G(V).$$

Consequences of an adjoint pair (f^*, f_*) :

▷ f^* is right exact; f_* left exact [true for abelian cats.]

▷ More generally, left adj. preserves colimits: cokernels, direct limits, stalk
right adj. preserves limits: kernels, products, pullback, inverse lims, ...

OVER $\rightarrow$

6 functor formalism on $D_c^b(X) := \text{[a framework across many different situations]}$

Def of $f_!$: For $s \in \Gamma(U, F)$, $\text{supp}(s) = \{x \in X \mid s(x) \neq 0\}$.

e.g. If $U = \text{Spec}(A)$ & $F = \tilde{M}$ is coherent, $s \in M$ has

$$\begin{aligned} \text{supp}(s) &= \{ \mathfrak{p} \in \text{Spec}(A) \mid s \neq 0 \text{ in } M_{\mathfrak{p}} \} \\ &= \overline{\{ \mathfrak{p} \mid \exists a \in A, a \notin \mathfrak{p}, a \cdot s = 0 \}} \cong \text{Spec}(A / \text{ann}_A(s)) \\ &\quad \Downarrow \\ &\quad \mathfrak{p} \supset \text{ann}_A(s) \end{aligned}$$

$\triangleright \Gamma_c(U, F) \subseteq \Gamma(U, F)$ is set of sections with compact support.

$\triangleright \Gamma_c$ remains left exact & so has a right derived functor $R\Gamma_c$ coh. $\rightarrow H_c^i(X, F^\bullet)$ or $H_c^i(X, F)$.] when $F = \underline{A}_X$, computes $H_c^i(X, A)$.

$\triangleright f_! F$ is given by:

$$f_! F(U) = \left\{ s \in \Gamma(f^{-1}(U), F) \mid f|_{\text{supp}(s)} : \text{supp}(s) \rightarrow U \text{ is proper} \right\}$$

supports are relatively cpt.

$f_! F \xrightarrow{\sim} f_* F$ isom. for proper F .

$\triangleright Rf_! F$ can be computed w/ "c-soft" shvs, i.e. those for which $\Gamma(X, F) \xrightarrow{\text{res}} \Gamma(K, F)$ is surjective $\forall$ compact $K \subset X$.
[Here, $\Gamma(K, F) := \varprojlim_{U \supset K} \Gamma(U, F)$]

$\triangleright f_!$ & $Rf_!$ satisfy base change w/o properness assumptions!

Thm (Base Change).

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ f' \downarrow & \lrcorner & \downarrow f \\ Y' & \xrightarrow{g'} & Y \end{array}$$

Cartesian. Then

$$\triangleright (g')^* \circ f_! = f'_! \circ g^* \text{ in } \text{Sh}(Y)$$

$$\triangleright (g')^* \circ Rf_! = Rf'_! \circ g^* \text{ in } D^b(Y)$$

e.g. $j: U \hookrightarrow X$ open
 $j_! F = \text{extension by zero}$

/ e.g. $A^2 - \{0\} \hookrightarrow A^2$
 $j_! \mathbb{Q} = \text{extn by zero} = j_! \mathbb{Q} = 0$
 $i_! \mathbb{Q} = \mathbb{Q} \otimes j_! \mathbb{Q} = \mathbb{Q}$

Def of $f^!$: — Very mysterious functor in general —

Hope: maybe $f_!$ has an adjoint functor $f^!$, i.e.

$$\text{Hom}(f_! F, G) \cong f_* \text{Hom}(F, f^! G)$$

NOTE: $A = (A_n)_{n \in \mathbb{Z}}$ on $\mathbb{Z} = \{\dots \circ \circ \dots\} \xrightarrow{f} \{pt\}$

$$f_* A = \prod_{n \in \mathbb{Z}} A_n, \quad f_! A = \bigoplus_{n \in \mathbb{Z}} A_n$$

Having a left adjoint $\Rightarrow$ left exact
in abelian cat $\Rightarrow$ preserves limits

$f_!$ cannot have left adjoint bc

$$\bigoplus_n \prod_k A \neq \prod_k \bigoplus_n A$$

Cannot exist in $\text{Sh}(X)$ also bc $f_!$ is not right exact.
But maybe it exists ~~in~~ in $D_c^b(X)$:

① $j: U \hookrightarrow X$: open inclusion

$$\begin{aligned} \Gamma(U, j^! G) &= \text{Hom}_U(\underline{A}_U, j^! G) = \Gamma(U, \text{Hom}(\underline{A}_U, j^! G)) \\ &= \Gamma(U, \text{Hom}(j_! \underline{A}_U, G)) = \text{Hom}(j^* j_! \underline{A}_U, j^* G) \\ &= \text{Hom}(\underline{A}_U, j^* G) = \Gamma(U, j^* G) \end{aligned}$$

$\Rightarrow j^! = j^*$ for open inclusions.

② $j: U \hookrightarrow X$; $f: X \rightarrow Y$

$$\begin{aligned} \Gamma(U, f^! G) &= \text{Hom}_U(\underline{A}_U, f^! G) = \text{Hom}_X(j_! \underline{A}_U, f^! G) \\ &= \Gamma(Y, f_* \text{Hom}(j_! \underline{A}_U, f^! G)) \\ &= \Gamma(Y, \text{Hom}(f_! j_! \underline{A}_U, G)) \\ &= \text{Hom}_Y(f_! j_! \underline{A}_U, G) \end{aligned}$$

$\Rightarrow f^! G$ should be $[U \mapsto \text{Hom}(f_! j_! \underline{A}_U, G)]$.

NOT a sheaf! $\longrightarrow$ replace ~~$\underline{A}_X$~~ with a resolⁿ
 $\underline{A}_X \longrightarrow K^\bullet$ [by c-soft sheaves]

$\Rightarrow$ Can ~~now~~ define $f^!$ on derived category!

Thm (Künneth). $R\Gamma_c(X_1 \times X_2, F_1 \boxtimes F_2) \cong R\Gamma_c(X_1, F_1) \boxtimes^L R\Gamma_c(X_2, F_2)$

Method for $f^! \mathcal{G}$:

① $\mathcal{G} \rightarrow \mathcal{I}^\bullet$ an injective resolution.

$\underline{A}_X \rightarrow K^\bullet$: flat c-soft resⁿ.

If $\mathcal{I}$ is injective, $f_K^! \mathcal{I}(U) = \text{Hom}_Y(f_! K_U, \mathcal{I})$

$R: U \mapsto f_!(K^\bullet|_U)$ is a "cosheaf" valued in $\mathcal{S}(Y)$

$$\S \quad f^! \mathcal{G} = \text{Hom}_{\mathcal{D}(Y)}(R, \mathcal{G})$$

Simple Cases:

- ① If $j: U \hookrightarrow X$ is open, then $j^! \cong j^*$ [exact]
- ② If $j: Z \hookrightarrow X$ is clsd, then $j^!$ is defined on sheaves $\S$
- $$j^! \mathcal{G}(U \cap Z) = \Gamma_{U \cap Z}(\mathcal{G})$$

supp contained in $U \cap Z$

③ Better defⁿ:

$$\begin{array}{ccc} & \overset{\circ}{X} & \\ \uparrow j & & \searrow p = \text{proper} \\ X & \xrightarrow{f} & Y \end{array} \Rightarrow f_! = p_* \circ j^!$$

Prop. $f^! = j^* \circ p^!$ (reduces to p : proper.)

for projective $X \xrightarrow{f} Y$

also have $f^! = i^! \pi^!$

$\downarrow i$ clsd $\rightarrow \mathbb{P}_Y^n \xrightarrow{\pi} Y$ projection

$$\mathbb{P}_Y^n = Y \times \mathbb{P}^n$$

$\triangleright i^!$ is done from before

$\triangleright \pi$ is controlled by "dualizing $\mathcal{O}_X$ " : $\pi^! \mathcal{G} \cong \mathcal{G} \boxtimes \omega_{Y/\mathbb{P}^n}$

$$\cong A_{\mathbb{P}^n}[2n]$$

④ $f: X \rightarrow Y$ smooth:

$$f^! \mathcal{F} = f^* \mathcal{F} \otimes \omega_{X/Y}[d], \quad \omega_X = p_X^! A_{\text{pt}}$$

constr. setting: $\omega_{X/Y} = \omega_{X/Y} = \omega_X \otimes_A f^*(\omega_Y)^\vee$ if $X \& S$
Y are smooth $A_X[2d]$

$$\omega_X \otimes f^* \omega_Y^\vee$$

$d = \text{rel. dim}$

étale: $f^! \mathcal{F} = f^* \mathcal{F}(d)[2d]$

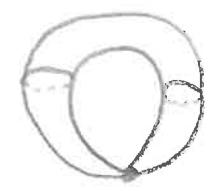
Check for a local chart $U \cong \mathbb{R}^n \rightarrow \{\text{pt}\}$ using adjunction.

(6)

Cor. $X = \text{pinched torus} = \{y^2 = x^3 - x^2\} \subset \mathbb{P}^2$ has

$$IH_2(X, \mathbb{Z}) = \begin{cases} \mathbb{Z} & i=0,2 \\ 0 & i=1 \end{cases}$$

Pf: $H_1(X - x, \mathbb{Z}) \cong \mathbb{Z} \xleftarrow{1} H_1(X, \mathbb{Z})$



bc gen. is a bdy in X

//

~~Return to this later~~

• Computing with chains is hard; often computed via normalizations.

Normalization: "a type of partial compactification"

Def. A domain A is normal if it is integrally closed in its field of fractions.

ie. $F = \text{Frac}(A) \supseteq A$. A is normal if for every

$a_1, \dots, a_m \in A$ & every $x \in F$ s.t. $x^m + a_1 x^{m-1} + \dots + a_m = 0$

we have $x \in A$.

Def. The normalizⁿ of A is

$\{x \in F \mid x \text{ satisfies } x^m + a_1 x^{m-1} + \dots + a_m = 0 \text{ for some } m \text{ & } a_i \in A\}$

② Thm (Stanley): For general rational P ,

$$h(P, t) = \sum_{k=0}^d \dim \underbrace{IH^{2k}(X_P, \mathbb{Q})}_{\text{"Intersection homology"}} \cdot t^k$$

~ We will return later in the course to prove this. ~

What is "Intersection (co)homology"?

A: provides a good (co)homology theory for singular varieties
"good": we can do intersection theory with it.

Intersection Theory for smooth varieties:

X : smooth, oriented n -manifold, $H_n(X; \mathbb{Z}) \cong \mathbb{Z} \ni [X]$ fundamental class

Thm (Poincaré Duality). $\cap [X]: H^i(X, \mathbb{Z}) \rightarrow H_{n-i}(X, \mathbb{Z})$

So with rational coeffs:

$$\begin{array}{ccc} H^i(X, \mathbb{Q}) & \xrightarrow{\sim} & H_{n-i}(X, \mathbb{Q}) \\ \parallel \mathbb{Z} & \text{universal coeffs. thm} & \parallel \mathbb{Z} \\ H_i(X, \mathbb{Q})^* & & H^{n-i}(X, \mathbb{Q})^* \end{array}$$

Cor: $h_i = \dim_{\mathbb{Q}} H_i(X, \mathbb{Q}) = \dim_{\mathbb{Q}} H^{n-i}(X, \mathbb{Q})$ is a symmetric/palindromic sequence.

Better: can specify the isomorphism $H^{n-i}(X, \mathbb{Q}) \cong H^{n+i}(X, \mathbb{Q})$

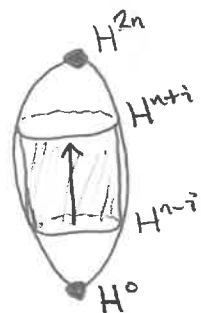
Assume $X \subseteq \mathbb{P}_{\mathbb{C}}^N$ is a complex, projective variety.

Choose a generic hyperplane $H \subset \mathbb{P}^N$

$$\leadsto [X \cap H] \in H_{2n-2}(X, \mathbb{Z}) \xleftrightarrow{\text{P.D.}} [H] \in H^2(X, \mathbb{Z})$$

Thm (Hard Lefschetz). $H^{n-i}(X, \mathbb{Q}) \xrightarrow[\sim]{-v[H]^i} H^{n+i}(X, \mathbb{Q})$ is isom.

↳ Rmk on how "hard" Hard Lefschetz is: hard enough Lefschetz didn't prove it in his 1924 paper; Hodge did in 1941!



Cor. h_i are also unimodal.